

EE147 Assignment #1 Questions

How many total thread blocks do we use?

- 1) The equation to the total thread blocks is $(\text{dim}/16) \times (\text{dim}/16)$. Therefore, with our default dimension of 1000, $1000/16$ is around 63. 63×63 is 3969. Therefore, there are 3969 total thread blocks.

Are all thread blocks full? That is, do all threads in the thread block have data to operate on?

- 2) No, not all thread blocks have data to operate on. We have edge cases in our gridDim equation. Therefore, the edge cases that do not fit into the if statement (`if(row < dim && col < dim)`) will not have data to work on because the index statement will not be executed. Cuda launches enough threads to cover a general region, but your actual matrix may not need the big of a size.

How can this basic Matrix Addition program be improved, i.e. what changes do you think can be made to speed up the code? Try out some of your ideas and note down your observations.

- 3) Some changes that can be made to speed up the code are changing the amount of threads in the block or reducing the number of calls to `cudaDeviceSynchronize()`. Neither of these changes made a significant difference in the time since the time the code was executed was already quick, but these are future techniques that could be used to optimize the speed of the program.